

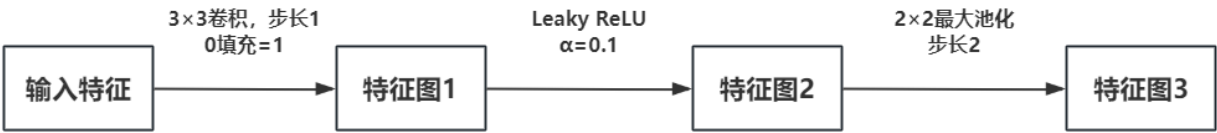
给定以下输入特征和卷积核参数，试计算特征图 1、2、3 的输出。

4	5	1	3	2	3
1	0	6	2	-1	2
4	7	-1	2	4	5
0	2	5	3	-1	1
1	1	0	-2	3	2
3	-1	4	1	2	1

输入特征

1	1	-1
1	-1	1
-1	1	1

卷积核



解：

1) 卷积：

输入特征填充后：

0	0	0	0	0	0	0	0
0	4	5	1	3	2	3	0
0	1	0	6	2	-1	2	0
0	4	7	-1	2	4	5	0
0	0	2	5	3	-1	1	0
0	1	1	0	-2	3	2	0
0	3	-1	4	1	2	1	0
0	0	0	0	0	0	0	0

卷积核：

1	1	-1
1	-1	1
-1	1	1

卷积运算计算特征图 1 各行元素：

第一行：

$$\begin{aligned} &0 \times 1 + 0 \times 1 + 0 \times (-1) + 0 \times 1 + 4 \times (-1) + 5 \times 1 + 0 \times (-1) + 1 \times 1 + 0 \times 1 = 2 \\ &0 \times 1 + 0 \times 1 + 0 \times (-1) + 4 \times 1 + 5 \times (-1) + 1 \times 1 + 1 \times (-1) + 0 \times 1 + 6 \times 1 = 5 \\ &0 \times 1 + 0 \times 1 + 0 \times (-1) + 5 \times 1 + 1 \times (-1) + 3 \times 1 + 0 \times (-1) + 6 \times 1 + 2 \times 1 = 15 \\ &0 \times 1 + 0 \times 1 + 0 \times (-1) + 1 \times 1 + 3 \times (-1) + 2 \times 1 + 6 \times (-1) + 2 \times 1 + (-1) \times 1 = -5 \\ &0 \times 1 + 0 \times 1 + 0 \times (-1) + 3 \times 1 + 2 \times (-1) + 3 \times 1 + 2 \times (-1) + (-1) \times 1 + 2 \times 1 = 3 \\ &0 \times 1 + 0 \times 1 + 0 \times (-1) + 2 \times 1 + 3 \times (-1) + 0 \times 1 + (-1) \times (-1) + 2 \times 1 + 0 \times 1 = 2 \end{aligned}$$

第二行:

$$\begin{aligned}0 \times 1 + 4 \times 1 + 5 \times (-1) + 0 \times 1 + 1 \times (-1) + 0 \times 1 + 0 \times (-1) + 4 \times 1 + 7 \times 1 &= 9 \\4 \times 1 + 5 \times 1 + 1 \times (-1) + 1 \times 1 + 0 \times (-1) + 6 \times 1 + 4 \times (-1) + 7 \times 1 + (-1) \times 1 &= 17 \\5 \times 1 + 1 \times 1 + 3 \times (-1) + 0 \times 1 + 6 \times (-1) + 2 \times 1 + 7 \times (-1) + (-1) \times 1 + 2 \times 1 &= -7 \\1 \times 1 + 3 \times 1 + 2 \times (-1) + 6 \times 1 + 2 \times (-1) + (-1) \times 1 + (-1) \times (-1) + 2 \times 1 + 4 \times 1 &= 12 \\3 \times 1 + 2 \times 1 + 3 \times (-1) + 2 \times 1 + (-1) \times (-1) + 2 \times 1 + 2 \times (-1) + 4 \times 1 + 5 \times 1 &= 14 \\2 \times 1 + 3 \times 1 + 0 \times (-1) + (-1) \times 1 + 2 \times (-1) + 0 \times 1 + 4 \times (-1) + 5 \times 1 + 0 \times 1 &= 3\end{aligned}$$

第三行:

$$\begin{aligned}0 \times 1 + 1 \times 1 + 0 \times (-1) + 0 \times 1 + 4 \times (-1) + 7 \times 1 + 0 \times (-1) + 0 \times 1 + 2 \times 1 &= 6 \\1 \times 1 + 0 \times 1 + 6 \times (-1) + 4 \times 1 + 7 \times (-1) + (-1) \times 1 + 0 \times (-1) + 2 \times 1 + 5 \times 1 &= (-2) \\0 \times 1 + 6 \times 1 + 2 \times (-1) + 7 \times 1 + (-1) \times (-1) + 2 \times 1 + 2 \times (-1) + 5 \times 1 + 3 \times 1 &= 20 \\6 \times 1 + 2 \times 1 + (-1) \times (-1) + (-1) \times 1 + 2 \times (-1) + 4 \times 1 + 5 \times (-1) + 3 \times 1 + (-1) \times 1 &= 7 \\2 \times 1 + (-1) \times 1 + 2 \times (-1) + 2 \times 1 + 4 \times (-1) + 5 \times 1 + 3 \times (-1) + (-1) \times 1 + 1 \times 1 &= (-1) \\(-1) \times 1 + 2 \times 1 + 0 \times (-1) + 4 \times 1 + 5 \times (-1) + 0 \times 1 + (-1) \times (-1) + 1 \times 1 + 0 \times 1 &= 2\end{aligned}$$

第四行:

$$\begin{aligned}0 \times 1 + 4 \times 1 + 7 \times (-1) + 0 \times 1 + 0 \times (-1) + 2 \times 1 + 0 \times (-1) + 1 \times 1 + 1 \times 1 &= 1 \\4 \times 1 + 7 \times 1 + (-1) \times (-1) + 0 \times 1 + 2 \times (-1) + 5 \times 1 + 1 \times (-1) + 1 \times 1 + 0 \times 1 &= 15 \\7 \times 1 + (-1) \times 1 + 2 \times (-1) + 2 \times 1 + 5 \times (-1) + 3 \times 1 + 1 \times (-1) + -0 \times 1 + (-2) \times 1 &= 1 \\(-1) \times 1 + 2 \times 1 + 4 \times (-1) + 5 \times 1 + 3 \times (-1) + (-1) \times 1 + 0 \times (-1) + (-2) \times 1 + 3 \times 1 &= (-1) \\2 \times 1 + 4 \times 1 + 5 \times (-1) + 3 \times 1 + (-1) \times (-1) + 1 \times 1 + (-2) \times (-1) + 3 \times 1 + 2 \times 1 &= 13 \\4 \times 1 + 5 \times 1 + 0 \times (-1) + (-1) \times 1 + 1 \times (-1) + 0 \times 1 + 3 \times (-1) + 2 \times 1 + 0 \times 1 &= 6\end{aligned}$$

第五行:

$$\begin{aligned}0 \times 1 + 0 \times 1 + 2 \times (-1) + 0 \times 1 + 1 \times (-1) + 1 \times 1 + 0 \times (-1) + 3 \times 1 + (-1) \times 1 &= 0 \\0 \times 1 + 2 \times 1 + 5 \times (-1) + 1 \times 1 + 1 \times (-1) + 0 \times 1 + 3 \times (-1) + (-1) \times 1 + 4 \times 1 &= -3 \\2 \times 1 + 5 \times 1 + 3 \times (-1) + 1 \times 1 + 0 \times (-1) + (-2) \times 1 + (-1) \times (-1) + 4 \times 1 + 1 \times 1 &= 9 \\5 \times 1 + 3 \times 1 + (-1) \times (-1) + 0 \times 1 + (-2) \times (-1) + 3 \times 1 + 4 \times (-1) + 1 \times 1 + 2 \times 1 &= 13 \\3 \times 1 + (-1) \times 1 + 1 \times (-1) + (-2) \times 1 + 3 \times (-1) + 2 \times 1 + 1 \times (-1) + 2 \times 1 + 1 \times 1 &= 0 \\(-1) \times 1 + 1 \times 1 + 0 \times (-1) + 3 \times 1 + 2 \times (-1) + 0 \times 1 + 2 \times (-1) + 1 \times 1 + 0 \times 1 &= 0\end{aligned}$$

第六行:

$$\begin{aligned}0 \times 1 + 1 \times 1 + 1 \times (-1) + 0 \times 1 + 3 \times (-1) + (-1) \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= -4 \\1 \times 1 + 1 \times 1 + 0 \times (-1) + 3 \times 1 + (-1) \times (-1) + 4 \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= 10 \\1 \times 1 + 0 \times 1 + (-2) \times (-1) + (-1) \times 1 + 4 \times (-1) + 1 \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= (-1) \\0 \times 1 + (-2) \times 1 + 3 \times (-1) + 4 \times 1 + 1 \times (-1) + 2 \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= 0 \\(-2) \times 1 + 3 \times 1 + 2 \times (-1) + 1 \times 1 + 2 \times (-1) + 1 \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= (-1) \\3 \times 1 + 2 \times 1 + 0 \times (-1) + 2 \times 1 + 1 \times (-1) + 0 \times 1 + 0 \times (-1) + 0 \times 1 + 0 \times 1 &= 6\end{aligned}$$

则特征图 1 为:

2	5	15	-5	3	2
9	17	-7	12	14	3
6	-2	20	7	-1	2
1	15	1	-1	13	6
0	-3	9	13	0	0
-4	10	-1	0	-1	6

2) 激活:

根据 Leaky ReLU 函数计算规则, 将特征图 1 中小于 0 的元素缩小 10 倍得特征图二如下:

2	5	15	-0.5	3	2
9	17	-0.7	12	14	3
6	-0.2	20	7	-0.1	2
1	15	1	-0.1	13	6
0	-0.3	9	13	0	0
-0.4	10	-0.1	0	-0.1	6

3) 池化:

依据最大池化规则取特征图二中相邻 2×2 区域内最大元素为特征图 3 的元素, 得特征图 3 如下:

17	15	14
15	20	13
10	13	6